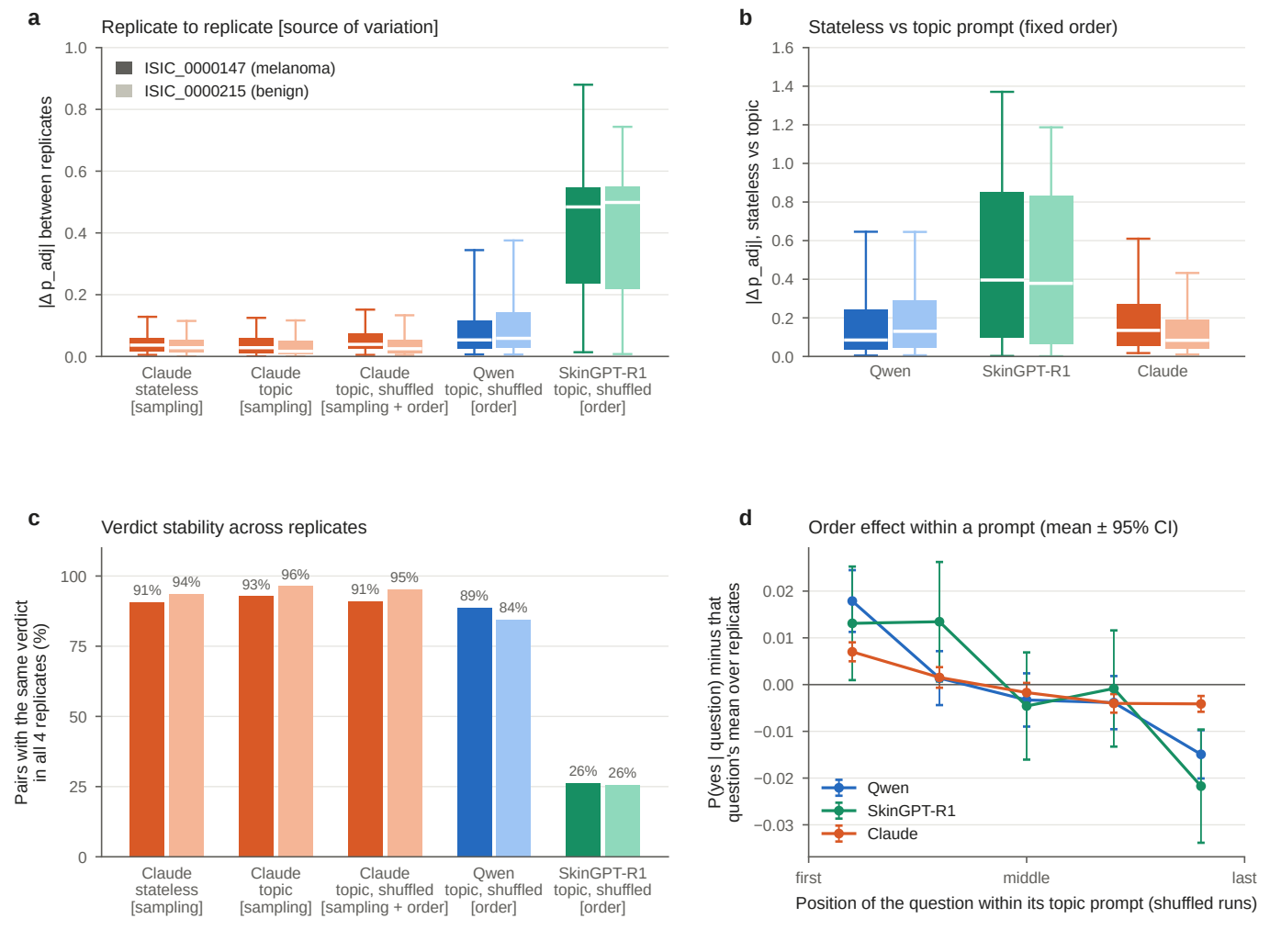
a
Replicate to replicate [source of variation]
ISIC_0000147 (melanoma)
ISIC_0000215 (benign)
|Δ p_adj| between replicates
1.0
0.8
0.6
0.4
0.2
0.0
Claude stateless [sampling]
Claude topic [sampling]
Claude topic, shuffled [sampling + order]
Qwen topic, shuffled [order]
SkinGPT-R1 topic, shuffled [order]
b
Stateless vs topic prompt (fixed order)
|Δ p_adj|, stateless vs topic
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Qwen
SkinGPT-R1
Claude
c
Verdict stability across replicates
Pairs with the same verdict in all 4 replicates (%)
100
75
50
25
0
91%
94%
93%
96%
91%
95%
89%
84%
26%
26%
Claude stateless [sampling]
Claude topic [sampling]
Claude topic, shuffled [sampling + order]
Qwen topic, shuffled [order]
SkinGPT-R1 topic, shuffled [order]
d
Order effect within a prompt (mean ± 95% CI)
P(yes | question) minus that question's mean over replicates
0.02
0.01
0.00
−0.01
−0.02
−0.03
Qwen
SkinGPT-R1
Claude
first
middle
last
Position of the question within its topic prompt (shuffled runs)